INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL PRACTICAL & LABORATORY MANUAL â¢ DEPARTMENT OF CSE

Laboratory Course: [BCS-353] Digital Electronics Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Prof. R. K. Patel (Dept. of CS**

Official Laboratory Manual: Digital Logic, Gates & Counter Design

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1. Laboratory Objectives & IC Pinout Verification

- Equipment: Digital IC Trainer Kit, Dual trace oscilloscope, TTL ICs (7400, 7402, 7404, 7408, 7432, 7486).
- Objectives: Practical design of combinational and sequential digital logic circuits.
- Rules: Verify VCC (+5V) and GND connections before powering trainer kit.

2. Prescribed Experiment Modules

- Experiment 1-3: Truth tables of universal gates, De Morgan’s laws verification, and K-Map logic minimization.
- Experiment 4-6: Multiplexer (74151) and Demultiplexer (74138) operational verification.
- Experiment 7-9: Flip-Flop conversion (SR to JK), Shift Registers, and Decade Synchronous Counter.